

Simplex IV: Column Geometry and Efficiency

What does a pivot mean geometrically, and what determines how much work simplex takes?

3.6 · Column geometry

3.7 · Computational efficiency

Lecture 8 · Bertsimas–Tsitsiklis §§3.6–3.7

Build on feasible bases, reduced costs, the ratio test, and the two-phase method from Simplex I–III.

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A new picture: columns are points, variables are weights

Previously, a point represented x . Here, each column A_j is a point and x_j is its weight.

$$Ax = \sum_j x_j A_j = b.$$

For a matrix A with m rows, study this mixture-form LP:

$$\min c^\top x \quad \text{s.t.} \quad Ax = b, \quad \sum_j x_j = 1, \quad x \geq 0.$$

Nonnegative weights summing to one make b a **weighted average** of the columns. Each column lists coefficients in the equations.

The sum-to-one condition belongs to this model. Adding it to an arbitrary LP can change the feasible set.

Self-study: reformulate a bounded feasible LP using scaled variables and a slack weight.

One five-column LP for the whole picture

Choose weights x_B, x_C, x_D, x_E, x_F . The labels B–F identify the five columns:

Column	B	C	D	E	F
Equation 1: A_{1j}	1/2	3	0	2	0
Equation 2: A_{2j}	1/2	0	3	1/2	0
Unit cost: c_j	7	4	4	1	4

$$\begin{aligned}
 \min \quad & z = 7x_B + 4x_C + 4x_D + x_E + 4x_F \\
 \text{s.t.} \quad & \sum_j x_j A_j = (1, 1)^\top, \quad \sum_j x_j = 1, \quad x_j \geq 0.
 \end{aligned}$$

The required weighted average is $b = (1, 1)$. For example, column C contributes 3 to equation 1, 0 to equation 2, and costs 4 per unit weight.

Add cost as the height of each column

Start with column C: $A_C = (3, 0)^\top$. Its two equation coefficients locate it on the floor.

Attach its unit cost $c_C = 4$ as a third coordinate:

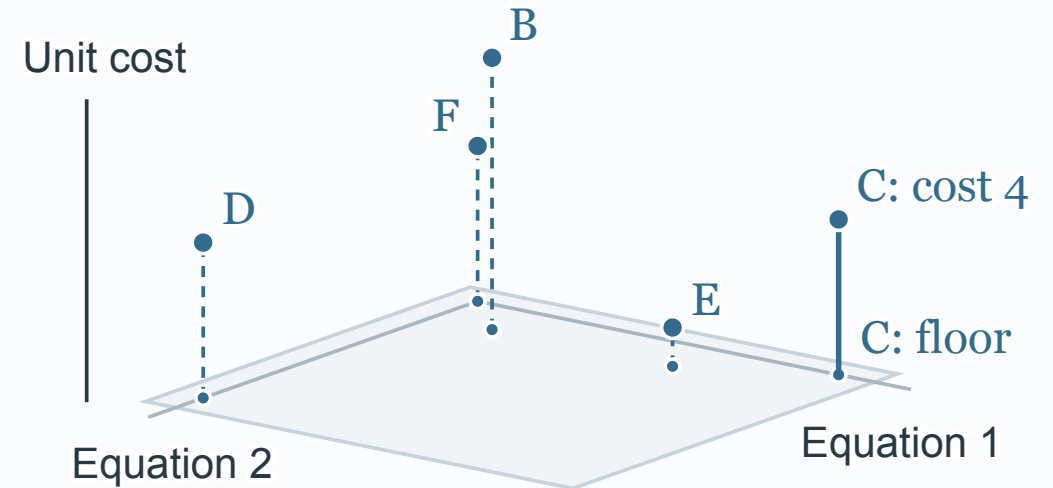
$$P_C = (3, 0, 4).$$

Do the same for all five columns:

$$P_j = \begin{pmatrix} A_j \\ c_j \end{pmatrix}.$$

The vertical segments show the attached costs.

The variables x_j will tell us how to mix these points.



A feasible mixture lands above the target

Give C, D, and F equal weights:

$$x_C = x_D = x_F = \frac{1}{3}, \quad x_B = x_E = 0.$$

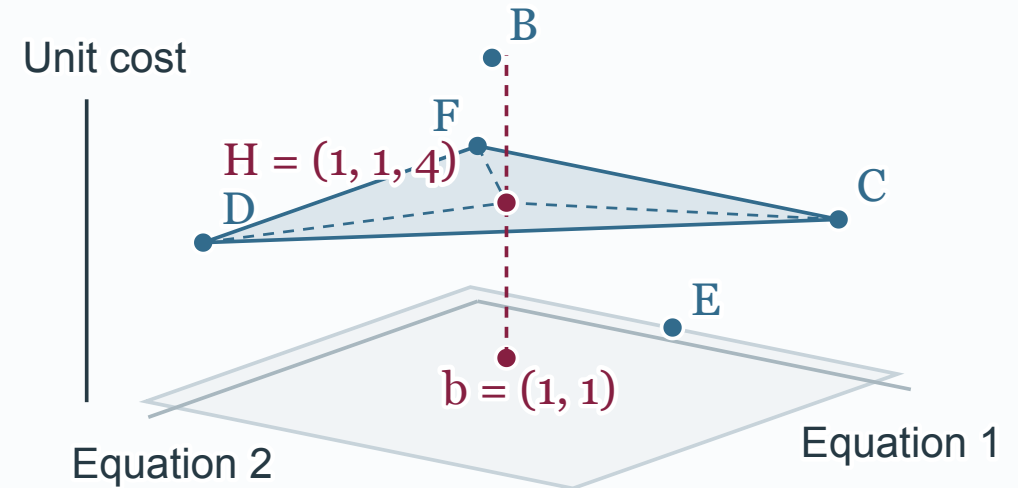
Their lifted points are $P_C = (3, 0, 4)$, $P_D = (0, 3, 4)$, and $P_F = (0, 0, 4)$.

$$\begin{aligned} H &= \frac{1}{3}P_C + \frac{1}{3}P_D + \frac{1}{3}P_F \\ &= (1, 1, 4). \end{aligned}$$

Horizontal coordinates: $b = (1, 1)$.

Height: objective value $z = 4$.

In general, the same weights give $\sum_j x_j P_j = (Ax, c^\top x) = (b, z)$.



Find the lowest attainable cost above the target

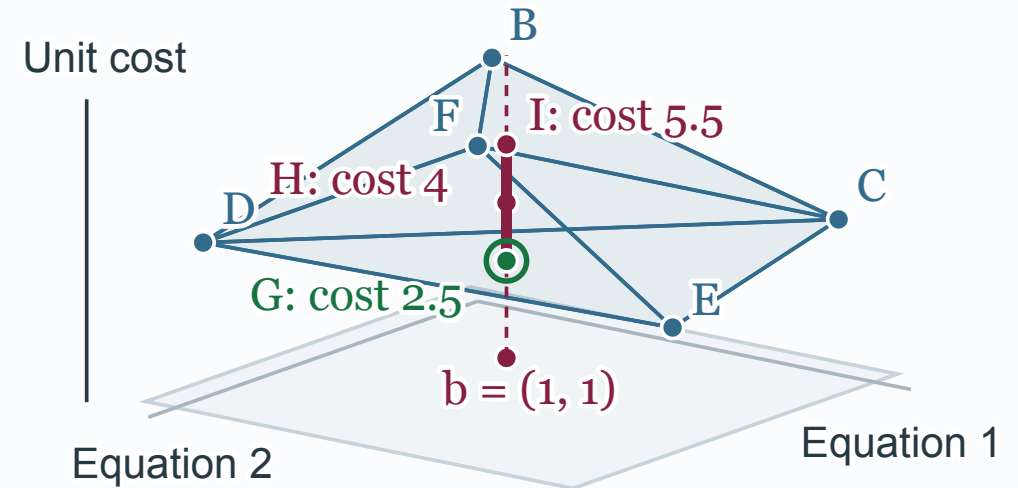
All nonnegative weighted averages with weights summing to one fill the convex hull of B–F.

The **requirement line** contains all $(1, 1, z)$: the two requirements stay fixed while cost varies.

Only the part of that line inside the hull is attainable:

$$\frac{5}{2} \leq z \leq \frac{11}{2}.$$

Minimization chooses the lowest point, G, with $z = 5/2$. H has cost 4; I has cost $11/2$.



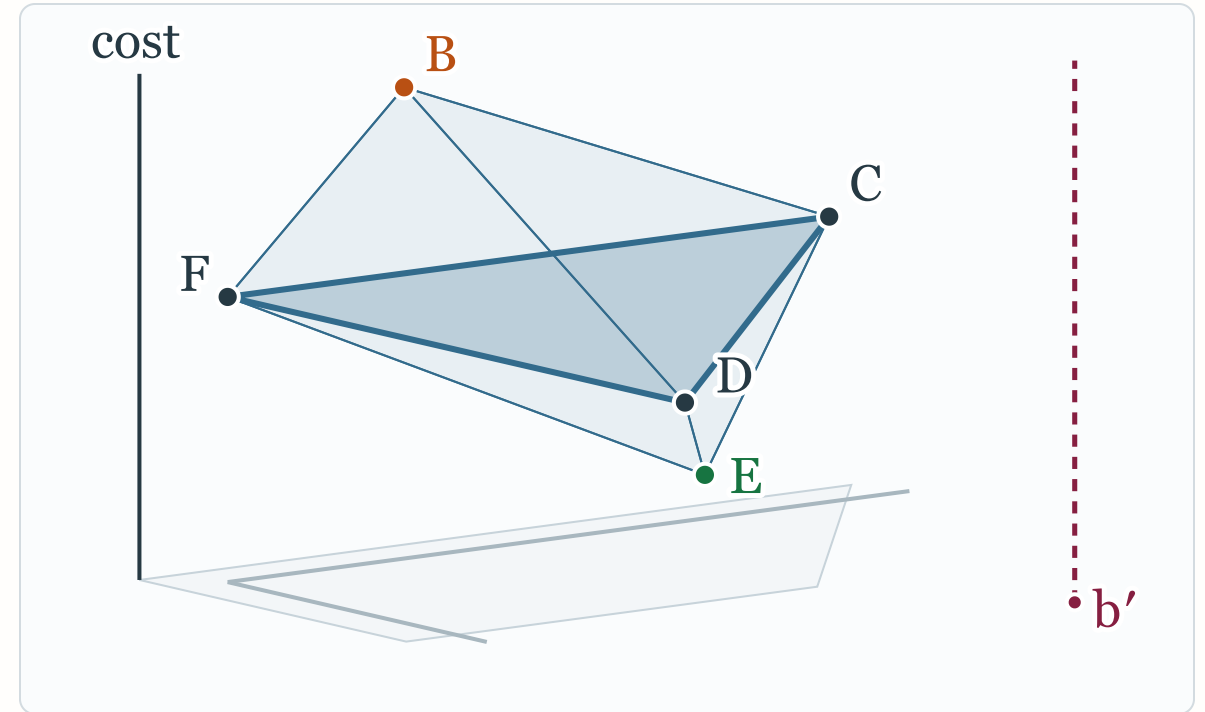
If the line misses the hull, no mixture is feasible

Keep the same five lifted columns, but change the requirement to $b' = (2.4, 2.4)$.

Every column satisfies $A_{1j} + A_{2j} \leq 3$.
Therefore every convex combination satisfies the same inequality.

$$b'_1 + b'_2 = 4.8 > 3.$$

The new requirement line never meets the hull.
The LP is infeasible, regardless of its objective.



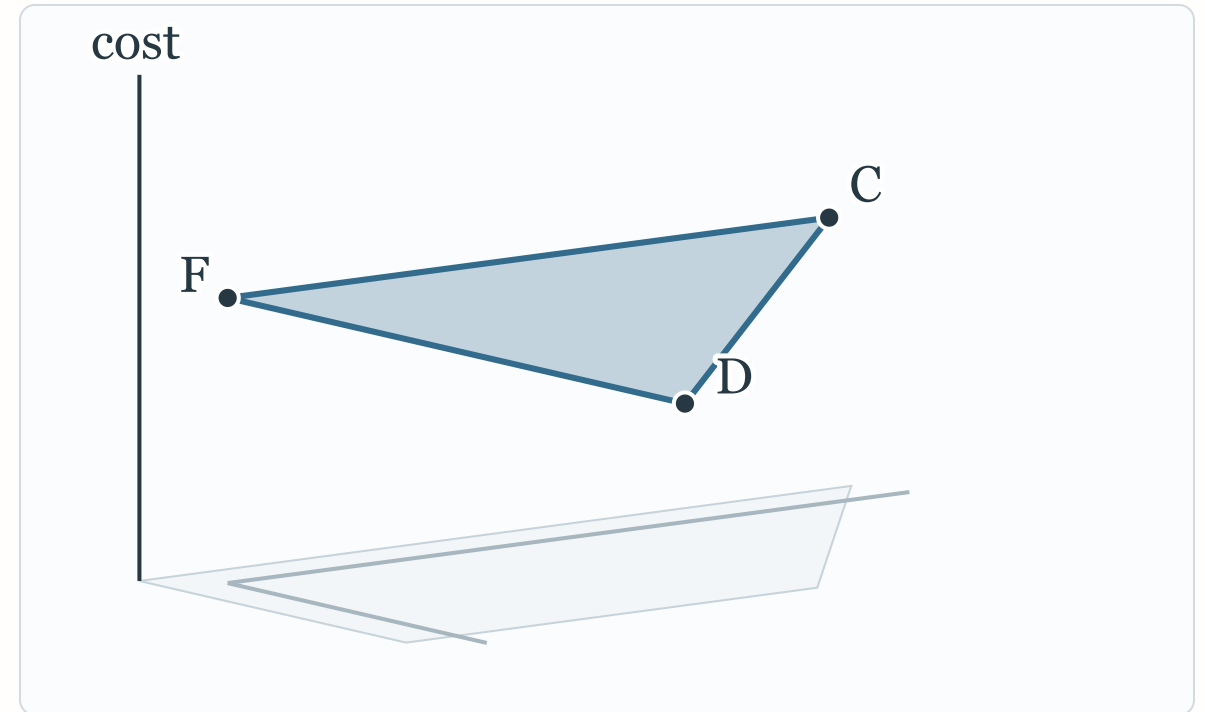
Affine independence gives a nonflat simplex

Points $v_1, \dots, v_{k+1} \in \mathbb{R}^n$ are **affinely independent** when

$$v_1 - v_{k+1}, \dots, v_k - v_{k+1}$$

are linearly independent. Necessarily $k \leq n$.

Their convex hull is a **k -dimensional simplex**.
Three noncollinear points give a triangle: CDF is a 2-simplex.



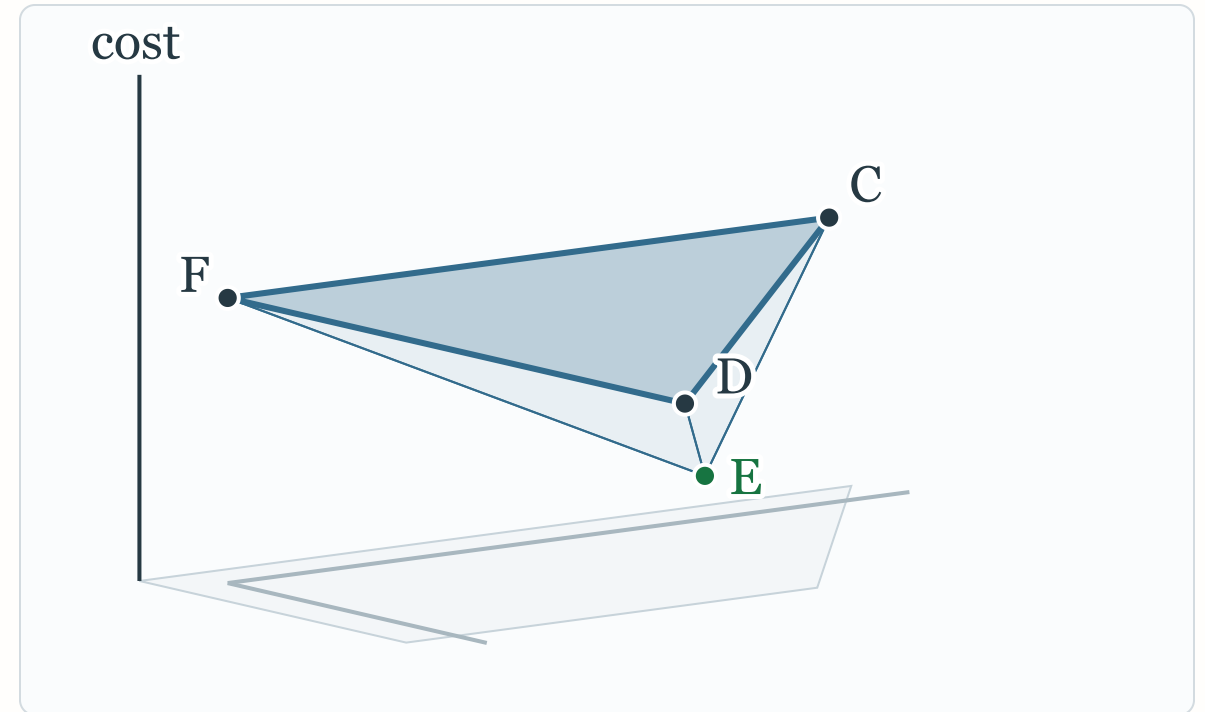
Four noncoplanar points give a tetrahedron

Four points are either coplanar or affinely independent.

In the independent case their convex hull is a 3-simplex: a tetrahedron, with four triangular faces.

C, D and F lie in one plane. E lies below it, so C, D, E and F are not coplanar.

Self-check: three differences from F span three dimensions. Which new direction does E contribute?



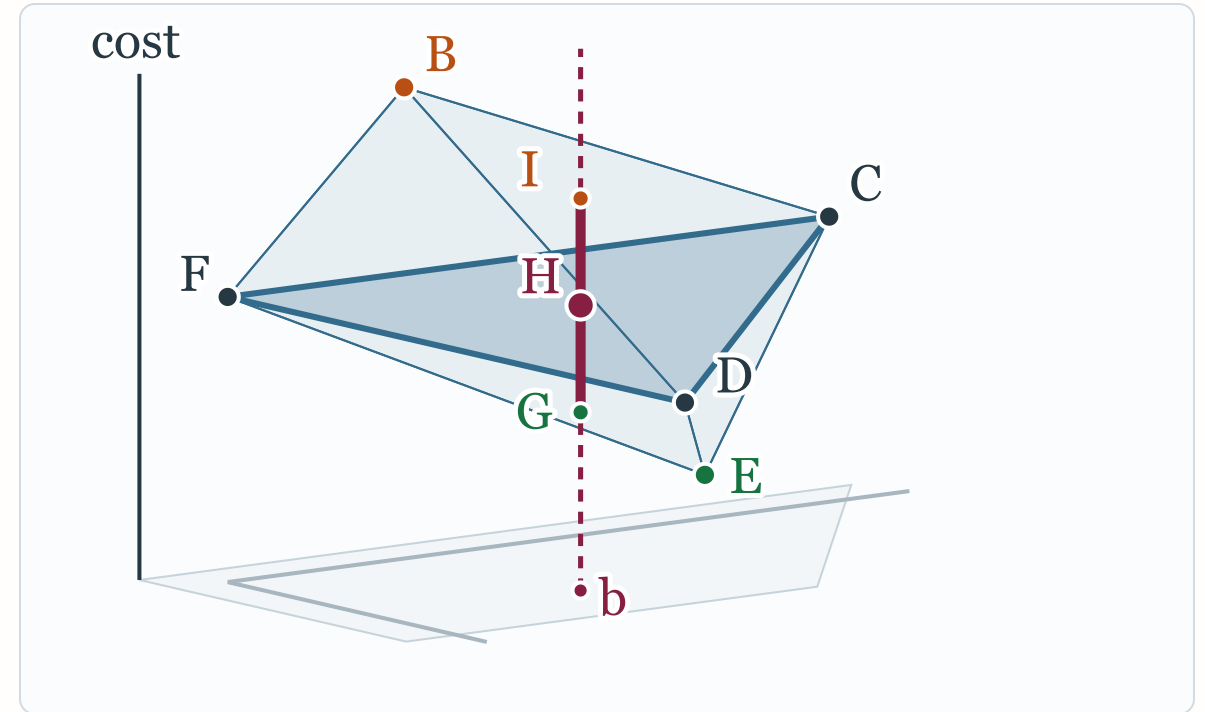
A basis selects three lifted points

Including convexity, the constraint matrix is

$$\tilde{A} = \begin{pmatrix} A \\ \mathbf{1}^\top \end{pmatrix}, \quad \tilde{b} = \begin{pmatrix} b \\ 1 \end{pmatrix}.$$

Assume $\tilde{A}$ has full row rank. A basis selects $m + 1$ independent columns $(A_j, 1)$.

Here $m = 2$: choose C, D, F. Their lifted versions are the **basic points**. B and E are nonbasic.

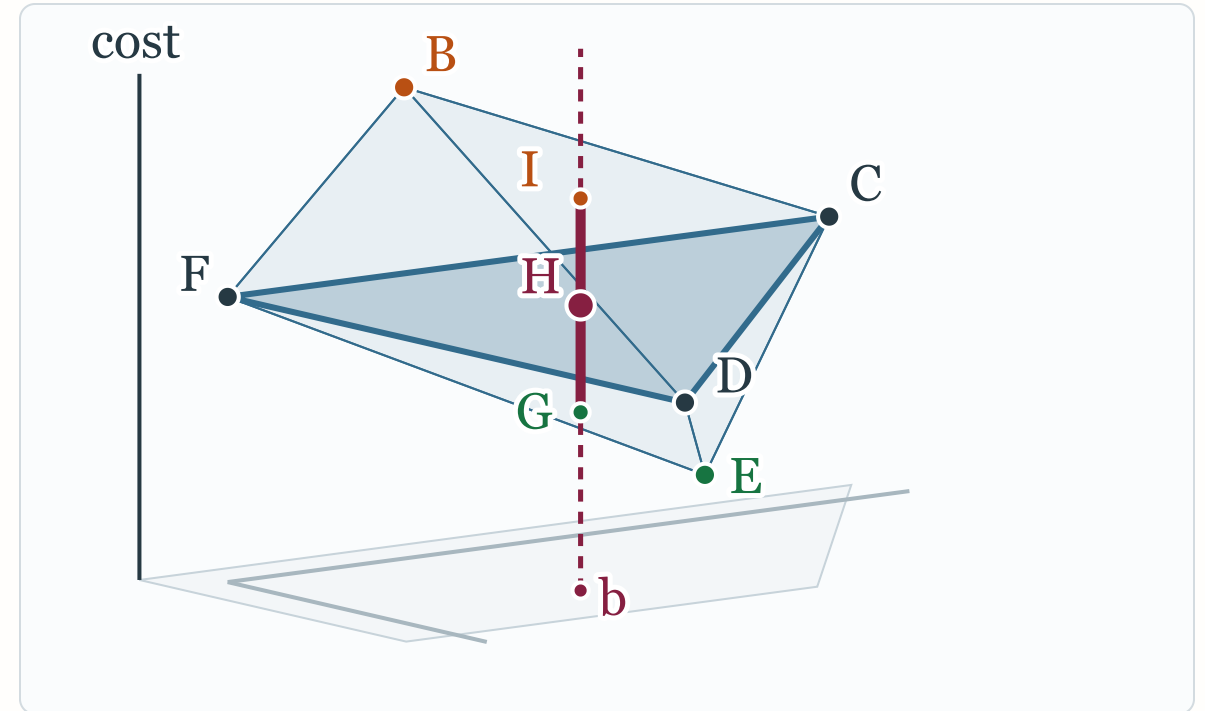


The selected points form the basic simplex

The basic columns $(A_j, 1)$ are independent, so their horizontal points A_j are affinely independent.

Lifting them to (A_j, c_j) preserves that affine independence: a relation between lifted differences would also be a relation between horizontal differences.

Their convex hull is the basic simplex. Our three basic points form the shaded triangle CDF.



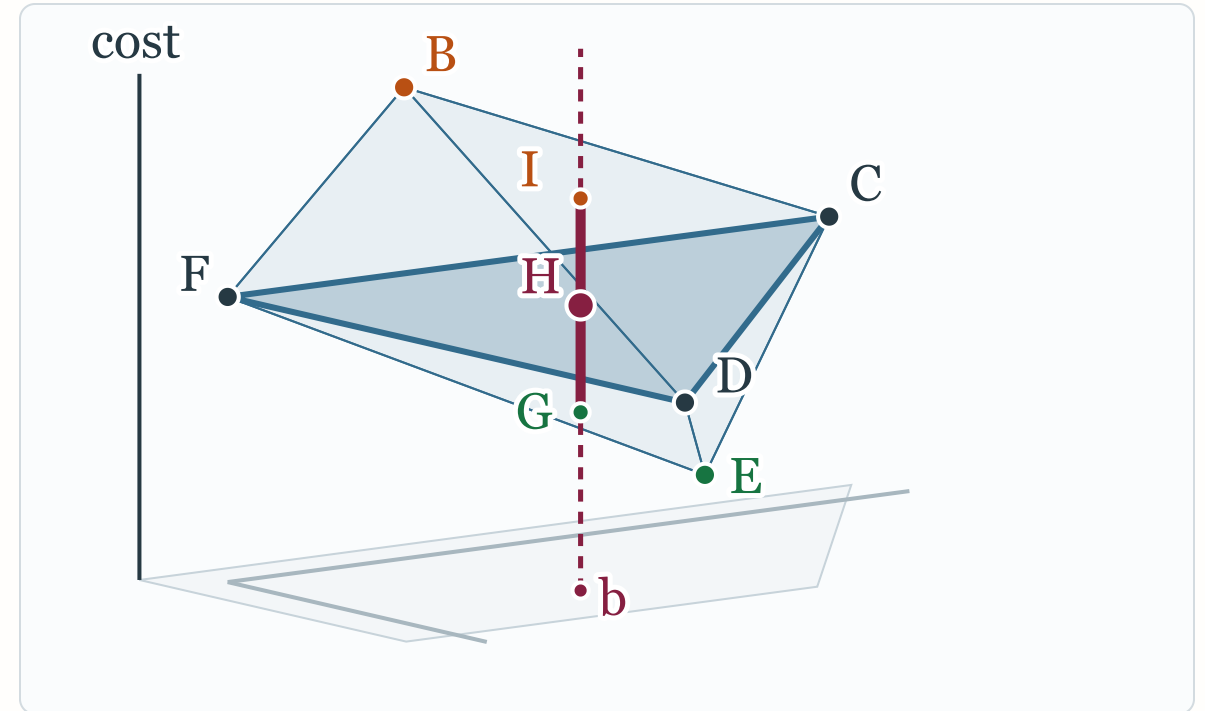
The intersection H determines the basic weights

Intersect the requirement line with triangle CDF.

$$H = \frac{1}{3}C + \frac{1}{3}D + \frac{1}{3}F = (1, 1, 4).$$

Thus $x_C = x_D = x_F = 1/3$, while $x_B = x_E = 0$. These are the entries of the **basic feasible solution**.

The point H is a lifted mixture. Its height 4 is the objective value; its coordinates are not the five decision variables.



A feasible basic mixture need not be optimal

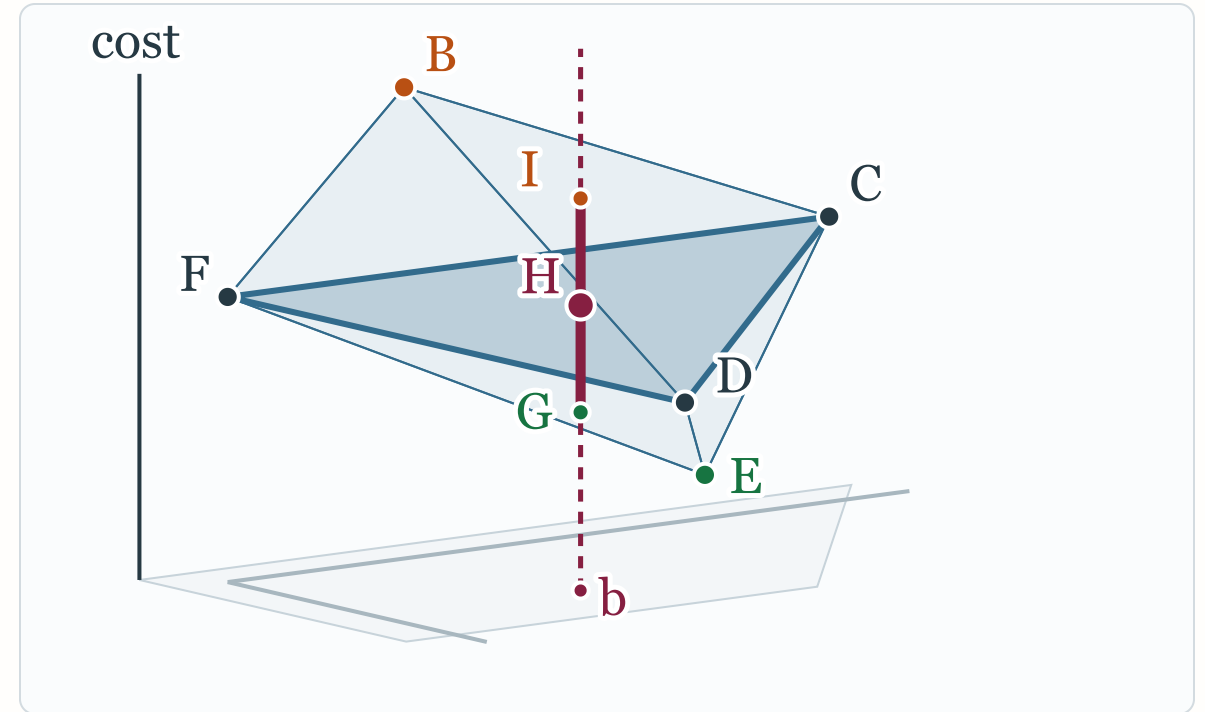
The same requirement line meets several possible basic triangles.

$$H = (1, 1, 4),$$

$$G = (1, 1, 5/2).$$

H uses CDF. The lower point G uses DEF. A change of basis can move the intersection downward while the requirement stays fixed.

The task is to choose a new basic point and an old basic point to remove.

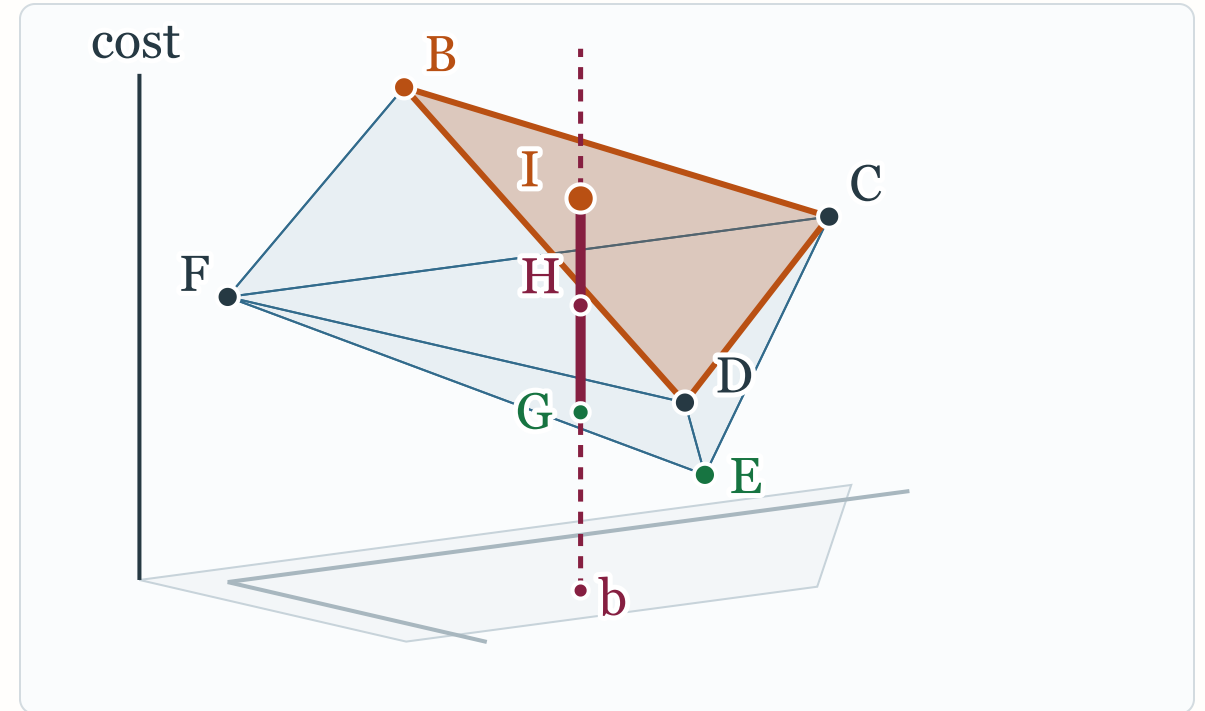


Candidate B: replace F and move upward to I

Starting with CDF, let B enter and F leave. The new basic simplex is BCD.

$$I = \frac{1}{2}B + \frac{1}{4}C + \frac{1}{4}D = (1, 1, 11/2).$$

The weights are feasible, but cost rises from 4 to $11/2$. This is a valid basis exchange with the wrong objective direction for minimization.

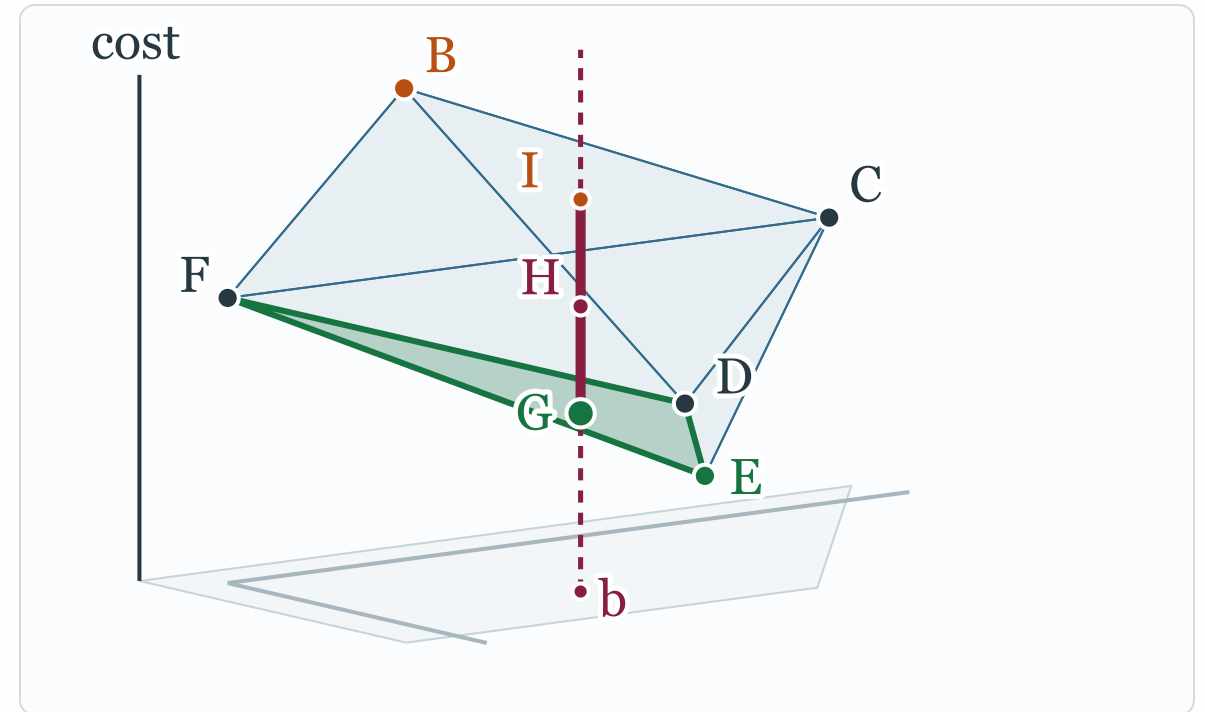


Candidate E: replace C and move downward to G

Instead, let E enter and C leave. The new basic simplex is DEF.

$$G = \frac{1}{4}D + \frac{1}{2}E + \frac{1}{4}F = (1, 1, 5/2).$$

Cost falls from 4 to 5/2. The next question is how to predict the better entering point before completing the exchange.



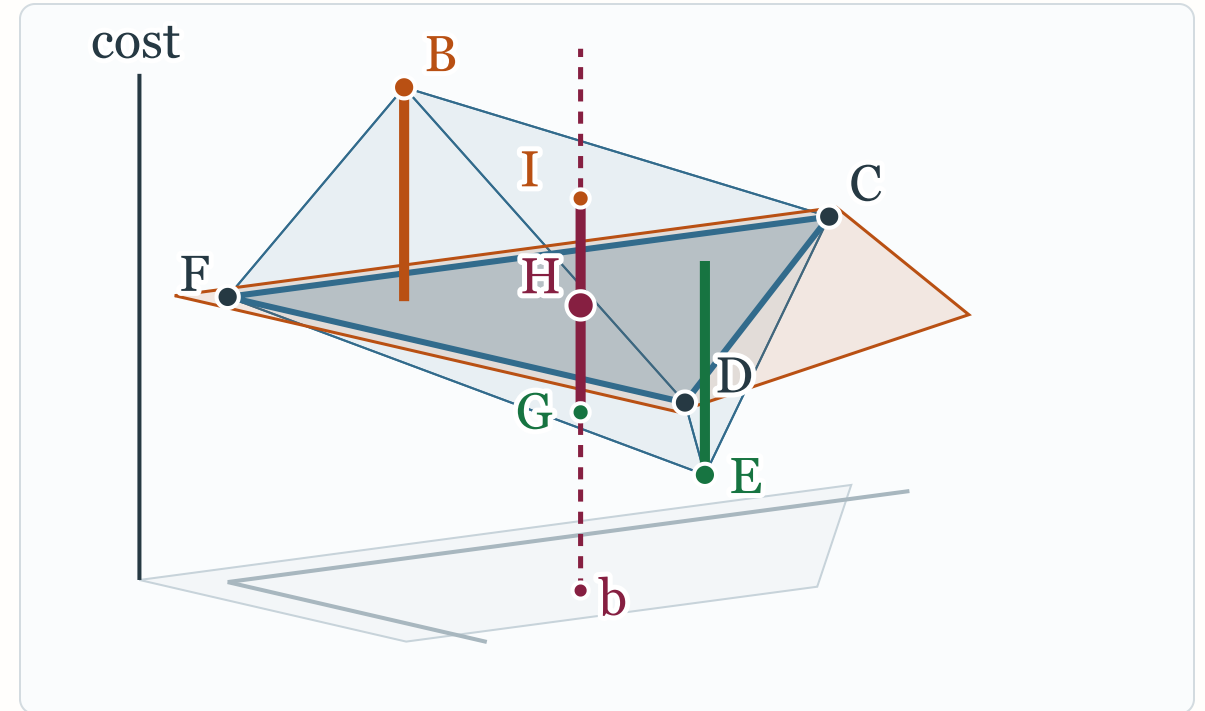
The dual plane predicts the objective slope

Extend the plane through the basic points C, D and F. This is the current dual plane.

$$z = 4.$$

E is below this plane; B is above it. Increasing the weight on E can decrease cost, while increasing the weight on B increases cost.

Strict improvement also needs a positive feasible step. A degenerate pivot may change the basis without changing the cost.



Reduced cost is a signed vertical gap

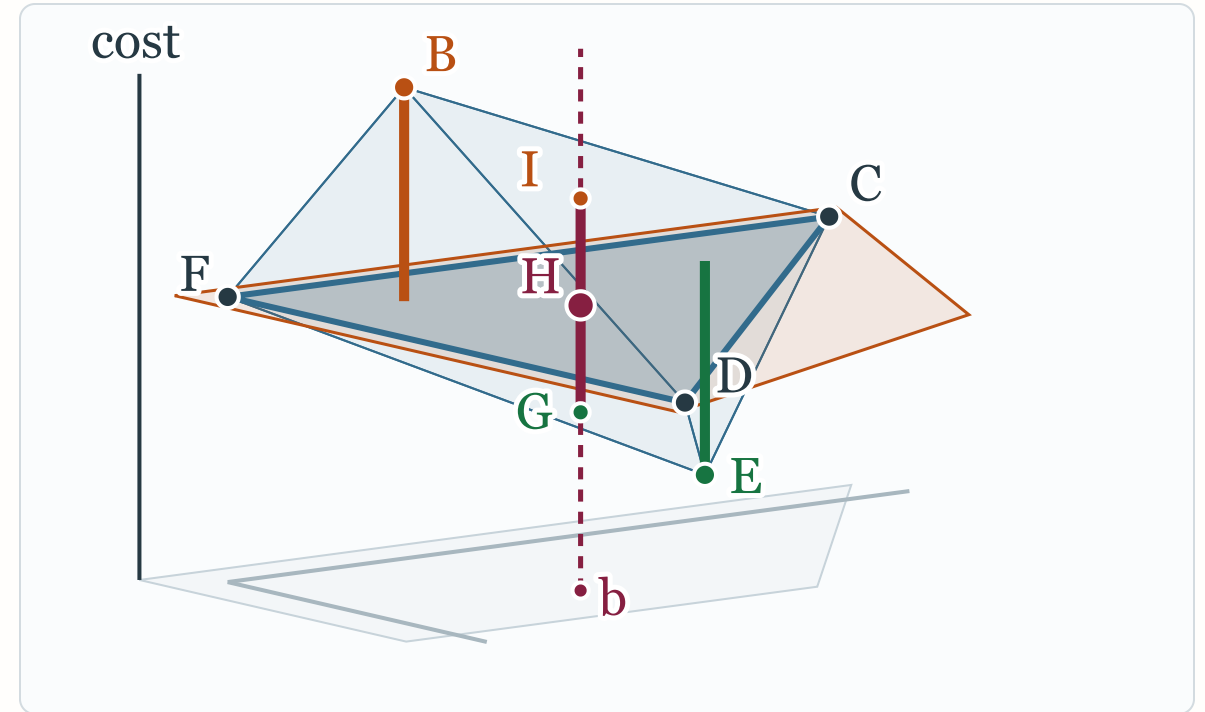
Write the dual plane as $z = \pi^\top a + \alpha$. It agrees with each basic point's cost.

$$\tilde{B}^\top \begin{pmatrix} \pi \\ \alpha \end{pmatrix} = c_B.$$

$$\bar{c}_j = c_j - (\pi^\top A_j + \alpha).$$

Subtract plane height from point height. Below the plane means a negative gap.

For CDF, $\pi = (0, 0)$, $\alpha = 4$: $\bar{c}_E = -3$, $\bar{c}_B = 3$.

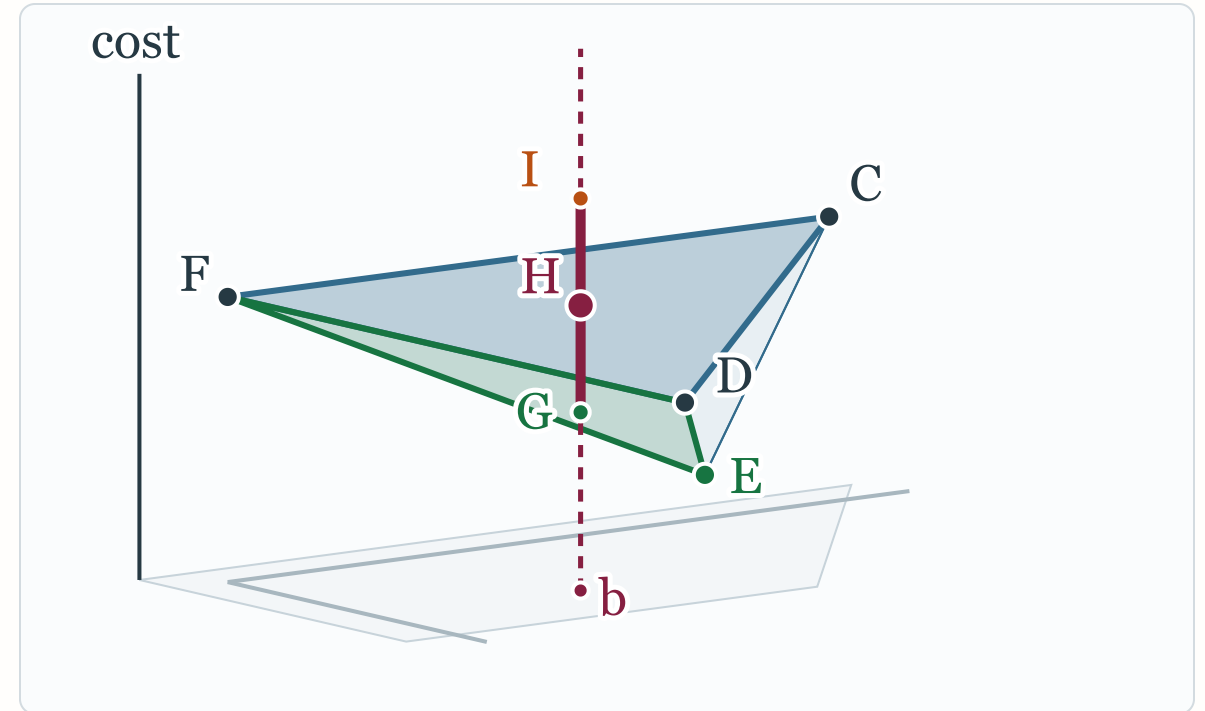


Adding E creates four possible boundary faces

The current triangle CDF together with E forms the tetrahedron CDEF.

Its four faces are CDF, CDE, CEF and DEF. The three new faces are candidates obtained by removing F, D or C.

In general an $(m + 1)$ -simplex has $m + 2$ faces: the old basic simplex plus $m + 1$ candidate replacements.

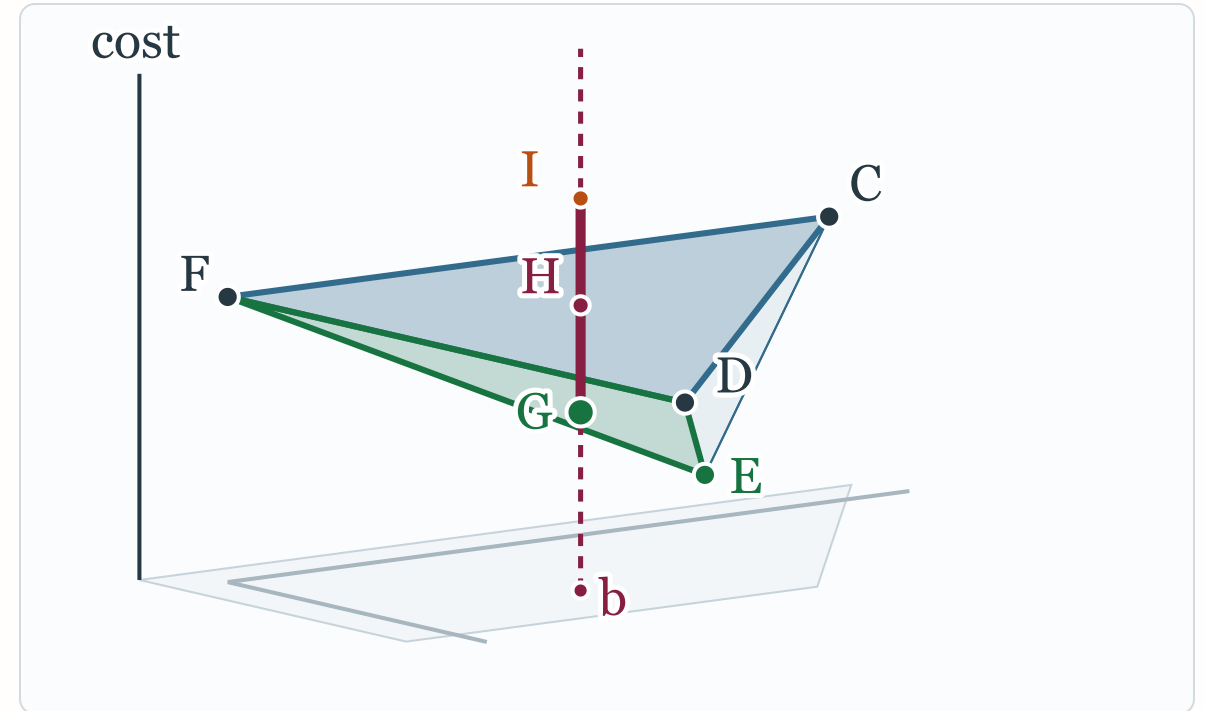


The requirement line selects the feasible new face

At H, the requirement line meets the old face CDF. Follow the line downward through the tetrahedron.

The next boundary intersection is G, on face DEF. That face contains E and excludes C: E enters and C leaves.

A candidate face that does not meet the requirement line cannot give nonnegative weights for this requirement.



The same face choice comes from the ratio test

Let $x_E = \theta$, starting from $x_C = x_D = x_F = 1/3$.
Preserving all three equalities gives

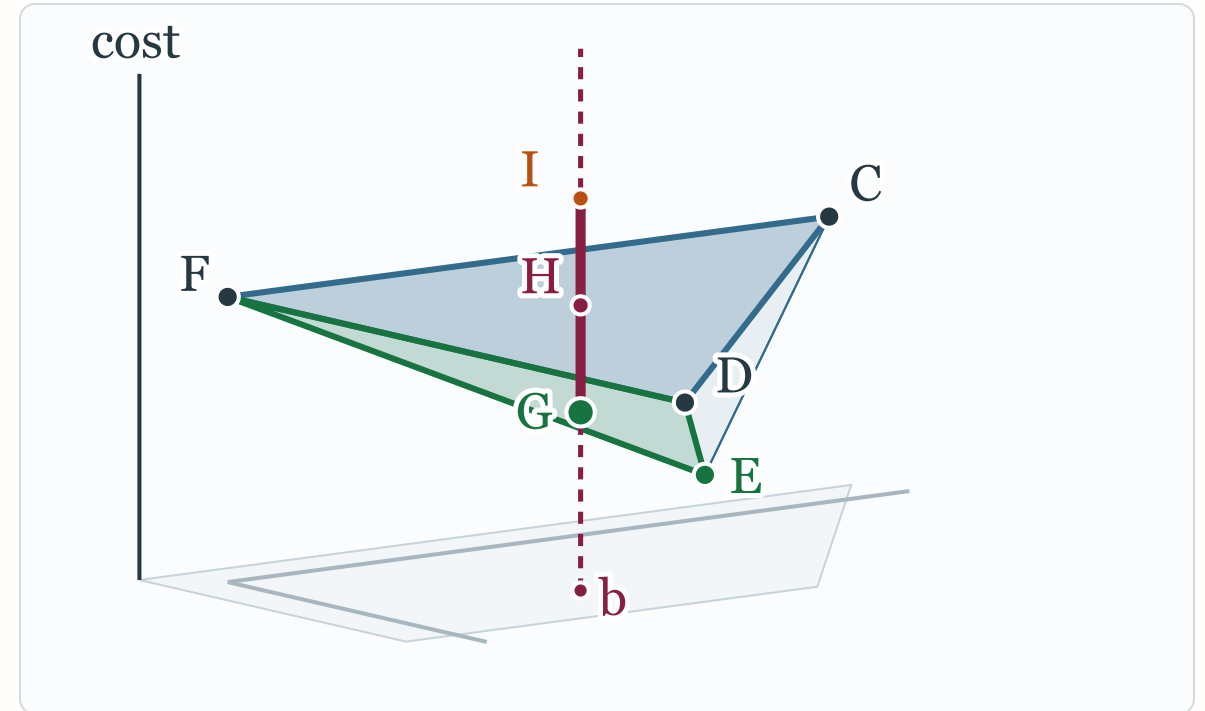
$$x_C = \frac{1}{3} - \frac{2}{3}\theta,$$

$$x_D = \frac{1}{3} - \frac{1}{6}\theta,$$

$$x_F = \frac{1}{3} - \frac{1}{6}\theta.$$

$$\theta^* = \min\{1/2, 2, 2\} = 1/2.$$

C reaches zero first, producing the face DEF.

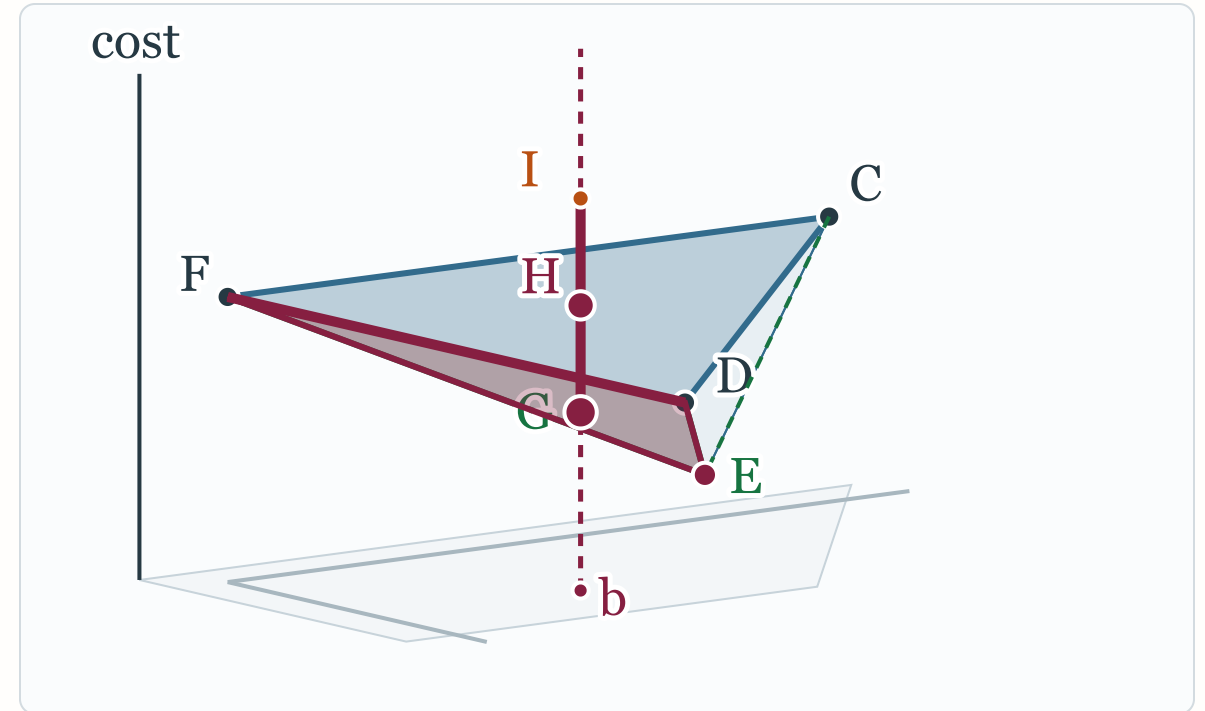


Pivot the triangle from CDF to DEF

The old triangle CDF and new triangle DEF share the hinge DF.

$$H = (1, 1, 4) \longrightarrow G = (1, 1, 5/2).$$

C leaves, E enters, and the cost decreases. The dashed corner path shows the physical analogy.



New triangle DEF; cost 2.50.

The final plane supports the entire convex hull

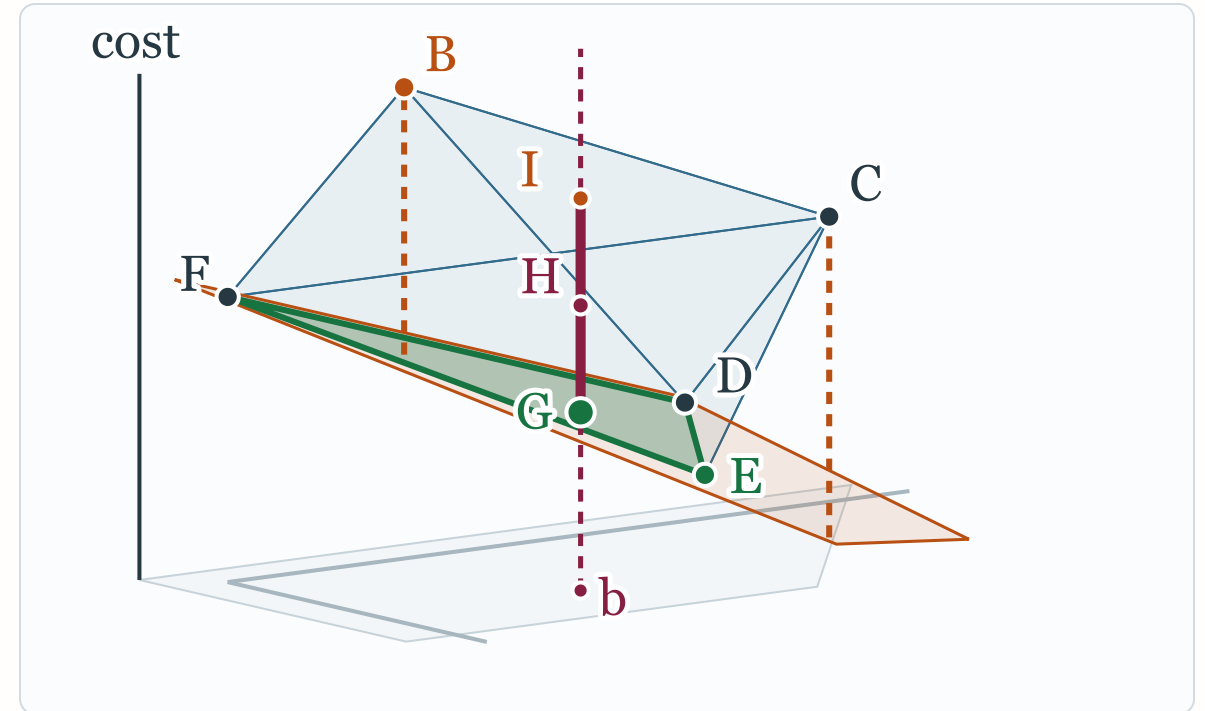
The plane through D, E and F is

$$z = 4 - \frac{3}{2}a_1.$$

Every lifted column lies on or above it. Therefore every convex combination satisfies $z \geq 4 - \frac{3}{2}a_1$.

At $b = (1, 1)$, this gives $z \geq 5/2$. The mixture G attains the bound, so it is optimal.

Self-study: verify the five signed gaps using the coordinate table.



Example 3.10: eight points, one horizontal coordinate

With $m = 1$, the basic simplex is a line segment. Use this constructed numerical realization of the eight-point picture:

j	1	2	3	4	5	6	7	8
A_j	5	1	-2	-3	3	4	2	-1
c_j	7	6	6	5	2	8	5	3

$$\min \sum_{j=1}^8 c_j x_j \quad \text{s.t.} \quad \sum_{j=1}^8 A_j x_j = 0, \quad \sum_{j=1}^8 x_j = 1, \quad x \geq 0.$$

We will choose the point with the most negative reduced cost: the largest downward vertical gap from the current dual line.

Initial basis 3, 6: point 5 has the largest downward gap

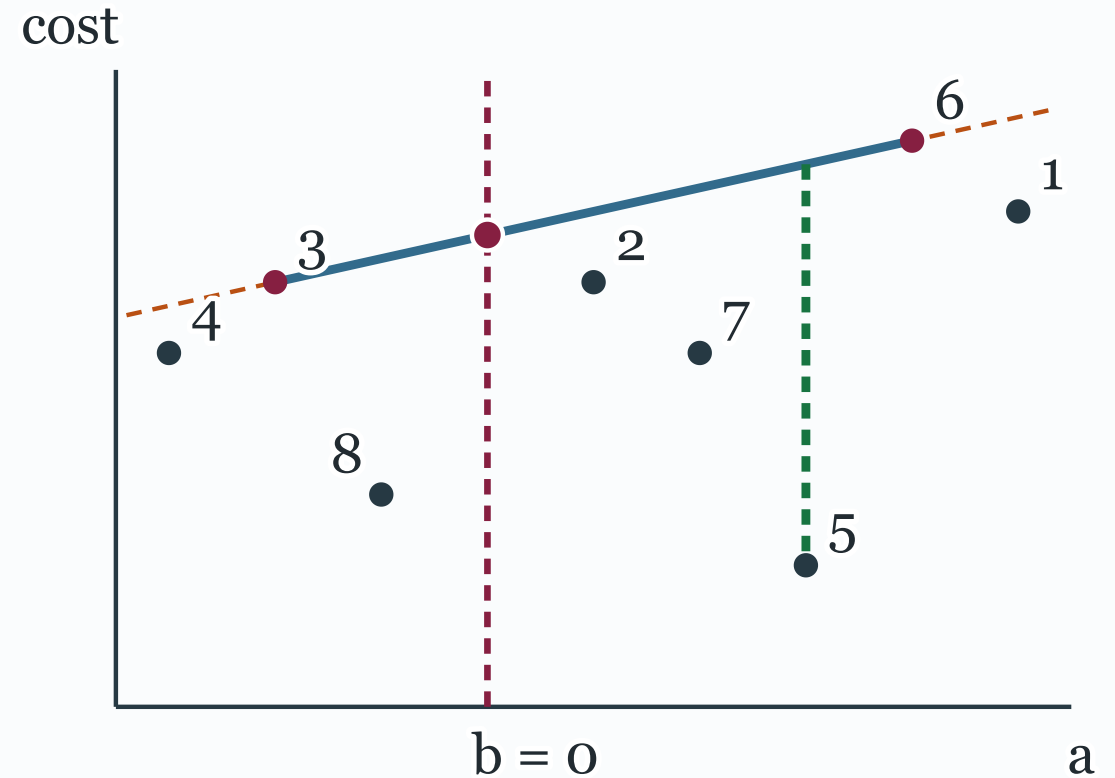
The line segment between points 3 and 6 crosses the requirement line.

$$x_3 = \frac{2}{3}, \quad x_6 = \frac{1}{3}, \quad z = \frac{20}{3}.$$

$$z = \frac{1}{3}a + \frac{20}{3}.$$

Point 5 has gap $2 - 23/3 = -17/3$, the most negative reduced cost.

Point 5 enters. The requirement lies between 3 and 5, so point 6 leaves.



Next basis 3, 5: point 8 is now the best candidate

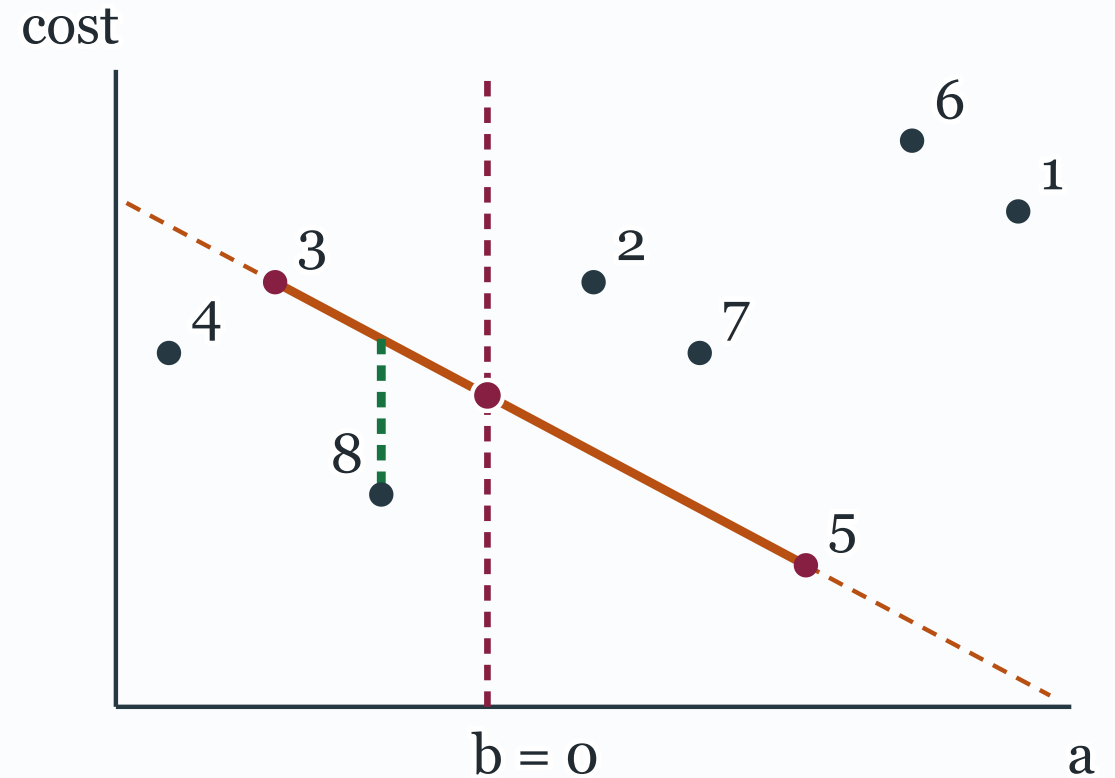
After the first pivot, the segment is 3, 5.

$$x_3 = \frac{3}{5}, \quad x_5 = \frac{2}{5}, \quad z = \frac{22}{5}.$$

$$z = -\frac{4}{5}a + \frac{22}{5}.$$

Now point 8 has the most negative gap: $3 - 26/5 = -11/5$.

Point 8 enters and point 3 leaves. The new segment must still cross the requirement line.



Final basis 5, 8: every point is on or above the line

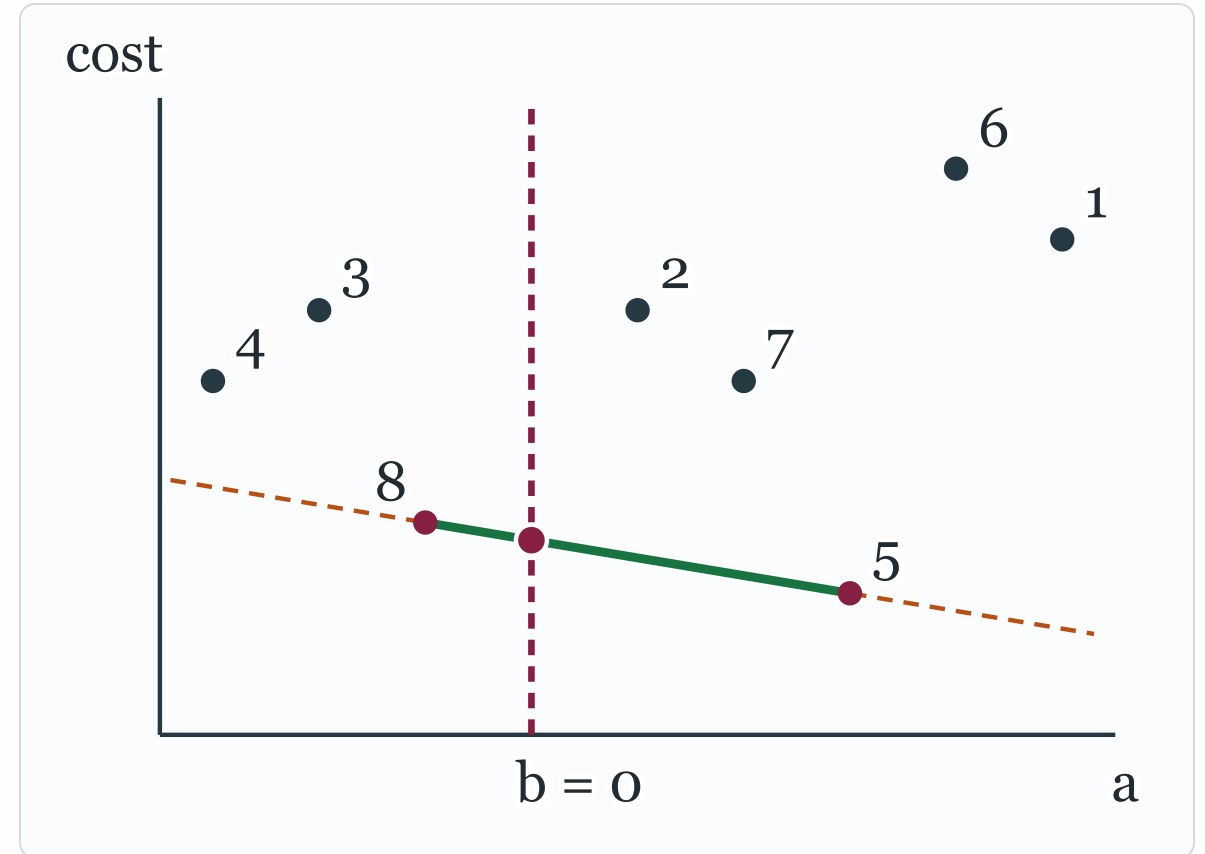
The final segment uses points 5 and 8.

$$x_5 = \frac{1}{4}, \quad x_8 = \frac{3}{4}, \quad z = \frac{11}{4}.$$

$$z = -\frac{1}{4}a + \frac{11}{4}.$$

All remaining points are above this line. Every nonbasic reduced cost is positive.

The current feasible mixture attains the lowest possible height at the requirement.



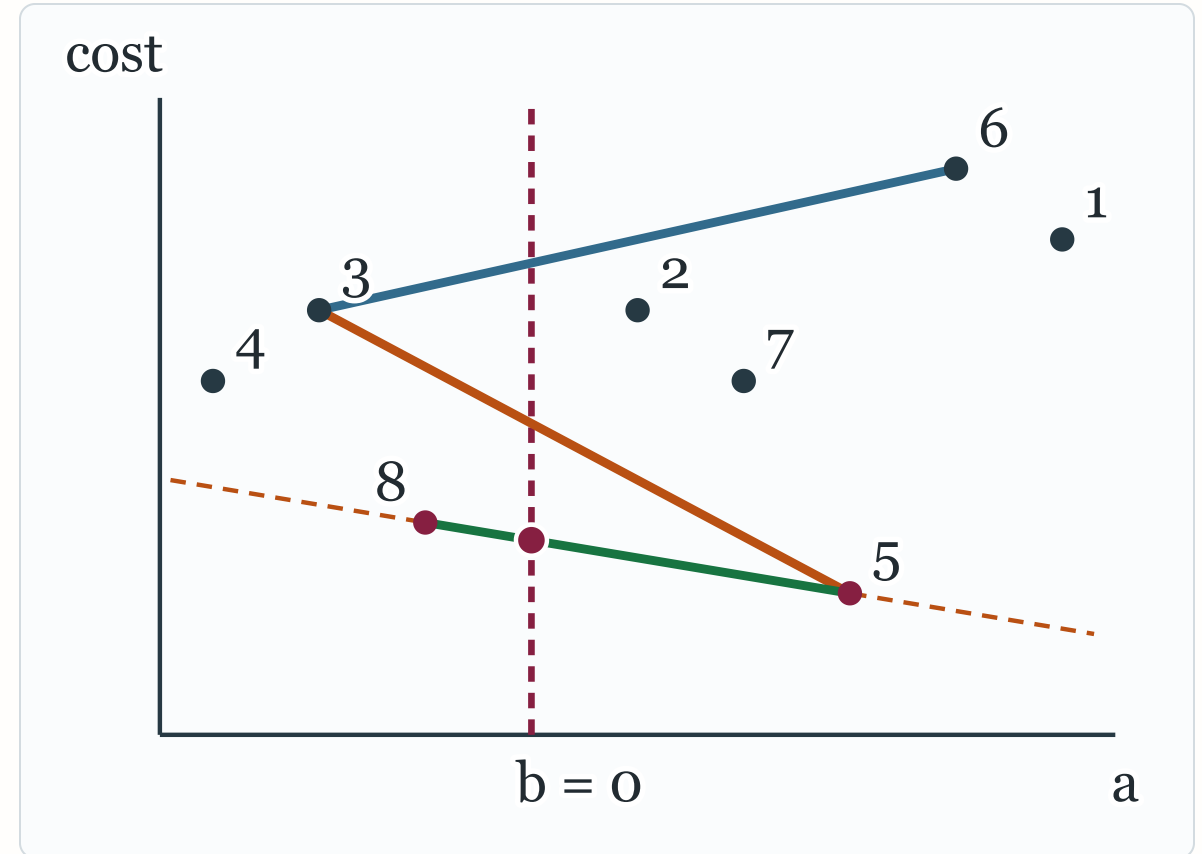
Read the entire path: 3, 6 → 3, 5 → 5, 8

Each pivot changes one endpoint of the basic segment while preserving its intersection with the requirement line.

$$\frac{20}{3} > \frac{22}{5} > \frac{11}{4}.$$

Self-study: recover the weights by solving the two equalities for each pair. Then calculate every reduced cost to confirm the entering choices.

The 2D segment rotates at a shared endpoint. The 3D triangle pivots around a shared edge. Both pictures describe the same basis exchange.



— initial: 3, 6 — next: 3, 5 — optimal: 5, 8

How much work does simplex require?

We now know how to start, pivot, and avoid cycling. Termination alone does not tell us how long the algorithm takes.

$$\text{total work} \approx \text{work per pivot} \times \text{number of pivots}.$$

We will examine both factors, then separate a deliberately difficult LP from typical behavior.

First factor: the work in one pivot

For standard form, let m be the number of independent equations and n the number of variables.

A full tableau has $O(mn)$ entries. One pivot updates them by row operations: $O(mn)$ arithmetic operations.

Revised simplex maintains and updates basis factors. Dense worst-case pricing and updates can also cost $O(mn)$, since $m \leq n$.

A fresh dense factorization costs $O(m^3)$. Sparsity matters; arithmetic counts also omit the bit length of exact numbers.

Bland's entering scan: compute reduced costs in increasing variable-index order; stop at the first negative one. If none is negative, check all nonbasic variables.

Why Lexicographic Comparison Needs More Than a Ratio

For basis B , write $\bar{b} = B^{-1}b$ and $\bar{A} = B^{-1}A$. Suppose x_j enters; consider a row with $\bar{a}_{ij} > 0$.

Ordinary test: compare one ratio. **Lex rule:** if those ratios tie, compare further entries in the fixed column order.

$$\frac{\bar{b}_i}{\bar{a}_{ij}} \quad \text{versus} \quad \frac{1}{\bar{a}_{ij}} \left[\bar{b}_i \mid \bar{a}_{i1}, \dots, \bar{a}_{in} \right].$$

Full tableau: store $[\bar{b} \mid \bar{A}]$. Read each candidate row and divide it by its positive entering-column coefficient.

Revised simplex: store basis factors for solving linear systems, instead of the full tableau. Obtain the needed normalized row for lex comparison by linear solves, without explicitly forming the inverse.

Second factor: how many pivots?

Many practical LPs need relatively few pivots. The reference describes counts roughly proportional to problem dimensions as an empirical observation.

That observation is not a bound for every LP. The feasible region can have exponentially many vertices.

To show slow behavior, we need more than many vertices: an **improving edge path** that a pivoting rule actually follows.

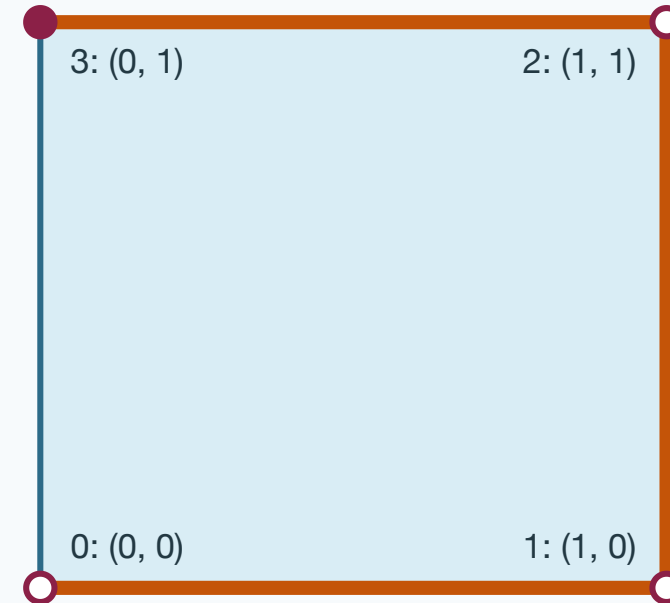
A cube already has exponentially many corners

Consider the unit cube:

$$0 \leq x_i \leq 1, \quad i = 1, \dots, n.$$

At a vertex each coordinate is either 0 or 1. There are 2^n binary choices.

A spanning path visits every vertex exactly once using edges.



minimize $f = -x_2$

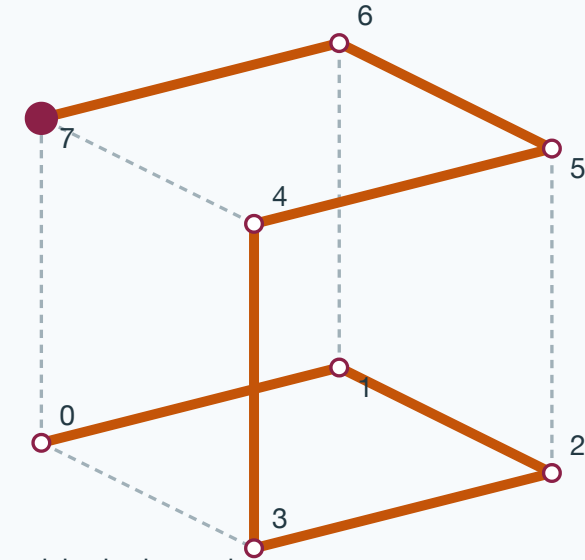
In two dimensions: 4 vertices and a path of 3 edges.

Build the path one dimension at a time

1. Follow the square path on $x_3 = 0$.
2. Cross one edge to $x_3 = 1$.
3. Follow the same square path in reverse.

$$L_n = 2L_{n-1} + 1 = 2^n - 1.$$

The same construction works recursively in every dimension.



orange: visited edges · ring: current vertex

Vertex 7 of 7; $x = (0, 0, 1)$; $f = -1$.

Why the ordinary cube is not enough

Choose the minimization objective $f(x) = -x_n$.

On the cube, half the vertices have $f = 0$, and half have $f = -1$.

Our spanning path has many edges along which the objective stays constant. It does not yet demonstrate a long sequence of *strictly improving* pivots.

Next we tilt the bounds, so movement in an earlier coordinate changes the possible value of the last coordinate.

Tilt the cube while keeping its combinatorics

Choose $0 < \epsilon < \frac{1}{2}$ and solve

$$\begin{aligned} \min \quad & -x_n \\ \text{s.t.} \quad & \epsilon \leq x_1 \leq 1, \\ & \epsilon x_{i-1} \leq x_i \leq 1 - \epsilon x_{i-1}, \quad i = 2, \dots, n. \end{aligned}$$

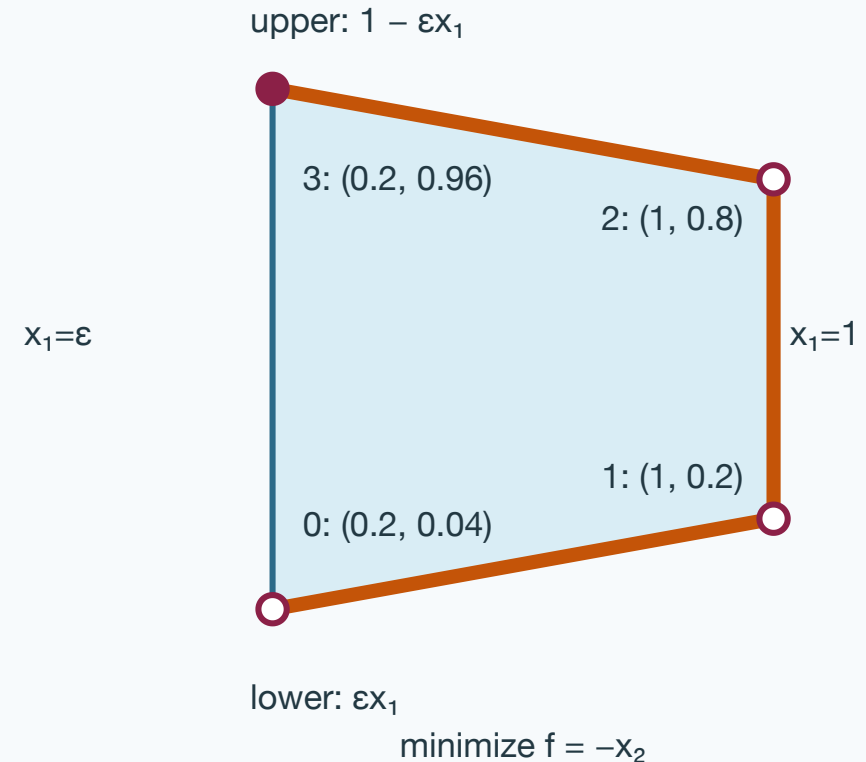
Each coordinate still has a lower and an upper bound. Their dependence on the previous coordinate makes the long path strictly improve the objective.

See the improvement in two dimensions

Set $\epsilon = 0.2$. Start at the vertex labeled 0 and follow the orange path.

vertex	0	1	2	3
x_2	.04	.20	.80	.96
$f = -x_2$	-.04	-.20	-.80	-.96

All three edge moves strictly decrease the objective. The fourth vertex is optimal.

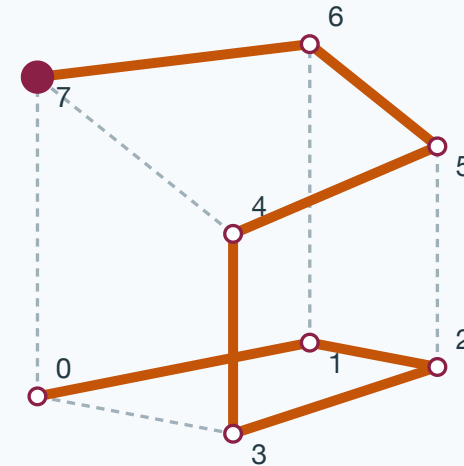


The same construction gives seven improving edges

Use the same inequalities with $n = 3$ and $\epsilon = 0.2$.

The figure shows the complete path from vertex 0 to vertex 7. The objective $f = -x_3$ decreases at every step.

The path visits all 8 vertices. In dimension n , it visits all 2^n vertices.



orange: visited edges · ring: current vertex

Vertex 7 of 7; $x = (0.2, 0.04, 0.992)$; $f = -0.992$.

Theorem 3.5: an exponential pivot path exists

For the perturbed cube just defined:

1. The feasible set has 2^n vertices.
2. They can be ordered so consecutive vertices share an edge and the cost strictly decreases.
3. A pivoting rule that follows this order takes $2^n - 1$ pivots.

This is an existence statement about a bad path and a rule that follows it. It does not say every rule follows that path.

Why there are exactly 2^n vertices

Choose a lower or upper bound for each coordinate, starting with x_1 .

The resulting triangular system has diagonal coefficients 1, so it defines a unique point.

Each lower bound is strictly below its upper bound: $\epsilon x_{i-1} < 1 - \epsilon x_{i-1}$, since $x_{i-1} \leq 1$ and $\epsilon < \frac{1}{2}$.

At most one bound per coordinate can be tight. A vertex needs n independent tight bounds, hence exactly one per coordinate. Every binary choice gives one distinct vertex.

Why reversing the upper path preserves improvement

On the lower face, $x_n = \epsilon x_{n-1}$: increasing x_{n-1} increases x_n .

Cross to the upper face, $x_n = 1 - \epsilon x_{n-1}$. This increases x_n , because $\epsilon x_{n-1} < 1 - \epsilon x_{n-1}$.

Then traverse the previous path backward. Now x_{n-1} decreases, so $1 - \epsilon x_{n-1}$ increases.

Consecutive vertices share $n - 1$ independent tight bounds. They are adjacent, and $-x_n$ strictly decreases throughout.

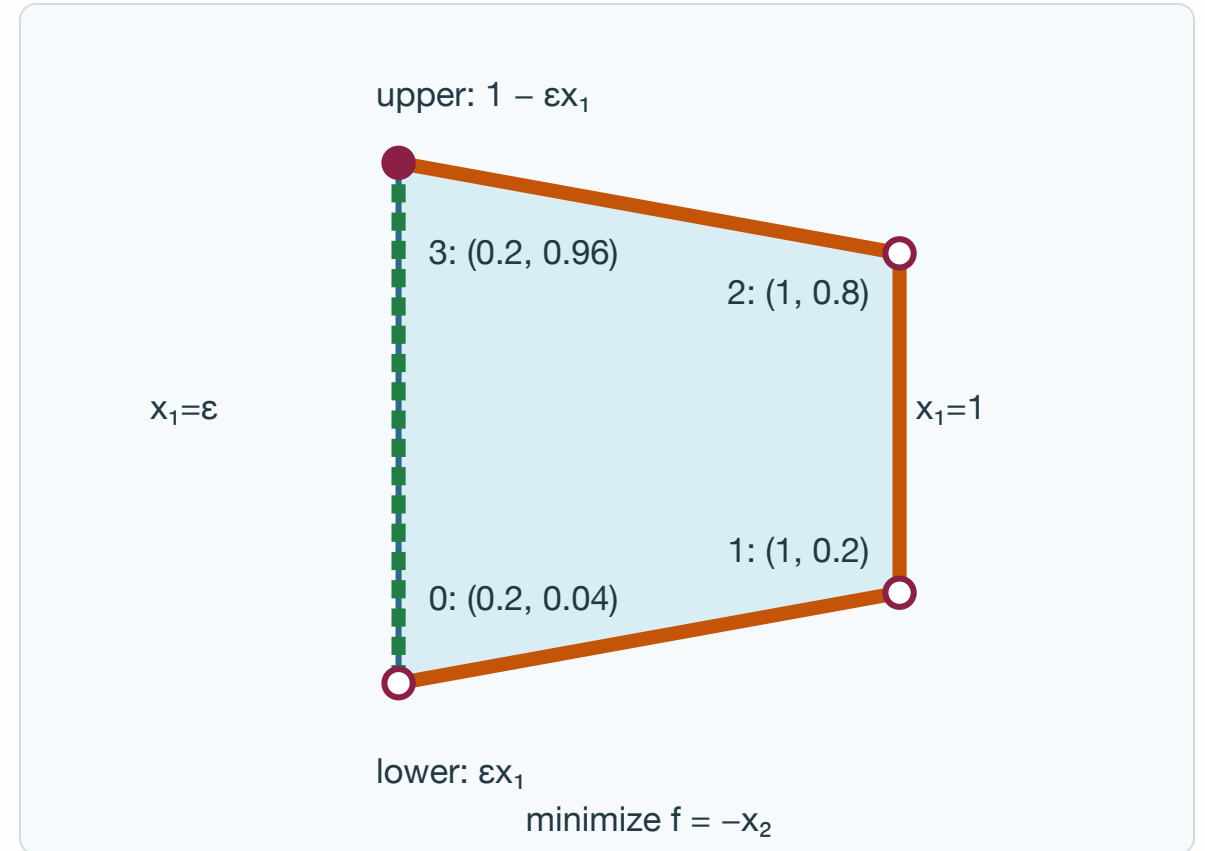
A different pivot choice can take a shortcut

The first and last vertices of this spanning path are adjacent.

Orange: follow the three-edge detour. Green: move directly to the optimum in one pivot.

The same first-to-last adjacency holds in the higher-dimensional construction.

Choosing a rule matters. A long available path does not force every algorithm to use it.



Can a pivoting rule always find a short route?

Exponential examples are known for many individual pivoting rules. That does not rule out every possible choice of rule.

The general search for a polynomial-time pivot rule remains a central question. A lower bound for one rule is not a lower bound for all rules.

We next separate two questions: **does a short path exist?** And **can an algorithm find an improving one efficiently?**

See [Disser and Mosis \(2025\)](#) for the scope of this open question.

Measure the shortest paths in the vertex graph

For a nonempty pointed polyhedron with vertices v, w , let $d(v, w)$ be the **minimum number of edges** in a path joining them.

$$D(P) = \max_{v, w \text{ vertices of } P} d(v, w).$$

This maximum of shortest-path distances is the **diameter** of the polyhedron's vertex-edge graph.

Distance counts edges, not Euclidean length. The paths in this definition need not improve an objective.

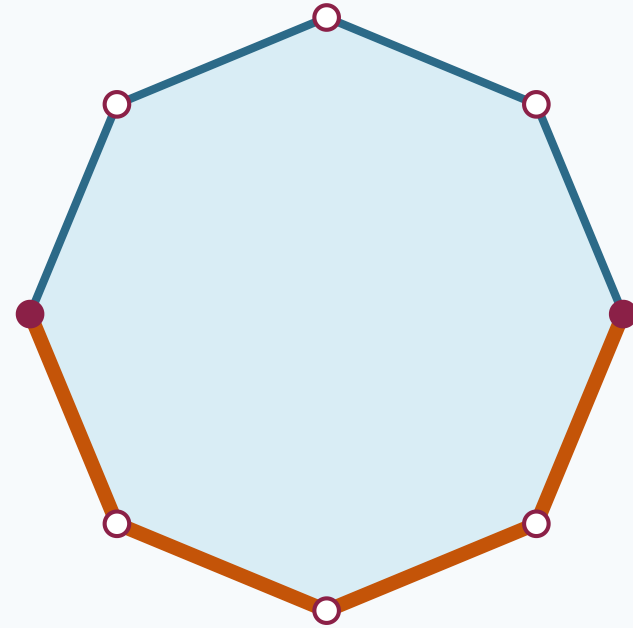
In two dimensions, a bounded polygon is a cycle

Here m counts inequality constraints and n is the ambient dimension. A bounded full-dimensional polygon has at most m edges.

Between two vertices, choose the shorter route around the boundary.

$$D(P) \leq \left\lfloor \frac{m}{2} \right\rfloor.$$

The illustrated octagon attains 4.



cycle: $D = 4 = \text{floor}(m / 2) \cdot m = 8$

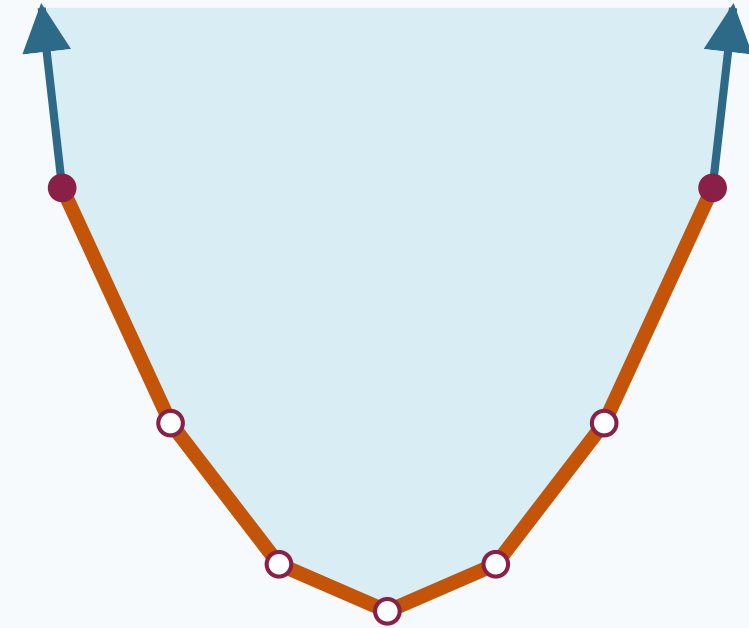
An unbounded polygon can have a longer chain

For a pointed, full-dimensional unbounded polygon, two boundary edges are rays. The finite vertices form a chain.

$$D(P) \leq m - 2.$$

With $m = 8$, the illustrated chain has seven vertices and six finite edges.

Thus the worst case over bounded and unbounded polygons is $\Delta(2, m) = m - 2$ for $m \geq 3$.



chain: $D = 6 = m - 2 \cdot m = 8$

Why diameter constrains every edge-following method

Suppose $d(v, w) = D(P)$. Start at v , and choose an objective for which w is the unique optimal vertex.

Every edge-following route from v to w needs at least $D(P)$ edge moves.

Define $\Delta(n, m)$ as the largest diameter among pointed n -dimensional polyhedra described by m inequalities.

A polynomial bound on diameter is necessary for a universally short edge path. It would not by itself tell us how to choose the pivots.

The Hirsch conjecture proposed a linear bound

The proposed bound was

$$\Delta(n, m) \leq m - n.$$

Klee and Walkup (1967) gave an unbounded-polyhedron counterexample.

Santos announced a bounded-polytope counterexample in 2010, published in 2012. The original Hirsch bound is therefore false even for polytopes.

[Santos, A counterexample to the Hirsch conjecture.](#)

A weaker question asks for a polynomial bound

The **polynomial Hirsch conjecture** asks whether $\Delta(n, m)$ is bounded by some polynomial in n and m .

A classical bound is

$$\Delta(n, m) \leq m^{1+\log_2 n} = (2n)^{\log_2 m}.$$

The exponent depends on the dimension. This is a quasipolynomial bound, not a polynomial bound jointly in both parameters.

The false linear conjecture and the weaker polynomial question must be kept separate.

A short path need not give a fast algorithm

Suppose a polynomial diameter bound were proved. Would that make simplex a polynomial-time algorithm?

No. Diameter promises some short edge path. It does not promise an improving path, identify the path, or bound the work needed to find it.

Degenerate pivots may also change the basis without traversing an edge to a new vertex. Their number needs its own control.

Worst-case behavior does not describe every instance

A worst-case result asks how badly a method can behave. An average-case result asks how it behaves under a specified distribution of inputs.

Many LPs encountered in practice require modest pivot counts. The reference describes roughly linear observed counts; these are observations, not universal theorems.

To turn “usually fast” into a mathematical claim, we must say what *usually* means.

Define the random experiment before taking an average

1. Fix the problem dimensions and choose a probability distribution over LP data.
2. Specify the algorithm, its initialization, and its pivoting rule.
3. Let T count the iterations and study $\mathbb{E}[T]$.

There is no single natural uniform distribution over all LPs. A result for one random model need not describe a different application.

One model: randomly orient fixed inequalities

Fix $c, a_1, \dots, a_m \in \mathbb{R}^n$ and $b_1, \dots, b_m$. Independently choose, with equal probability,

$$a_i^\top x \leq b_i \quad \text{or} \quad a_i^\top x \geq b_i.$$

There are 2^m sign choices. Suppose $L > 0$ produce feasible LPs.

The reference's Haimovich (1983) result bounds the average by $n/2$ pivots over those L feasible problems, for the specially specified pivoting procedure.

The distribution and procedure are essential assumptions; this is not a bound for arbitrary simplex implementations.

Smoothed analysis: perturb an arbitrary starting instance

Average-case analysis randomizes the whole input. **Smoothed analysis** starts with arbitrary data and adds small random perturbations.

For an appropriate shadow-vertex simplex algorithm, Gaussian perturbation models give polynomial expected complexity in the problem dimensions and $1/\sigma$, where σ measures the noise scale.

This helps explain why carefully constructed difficult instances can be fragile. It does not give every pivoting rule a polynomial worst-case guarantee.

Spielman and Teng, Smoothed Analysis (2004).

Connect the geometry to the computational work

Geometry explains the step

Replace one basic point. Reduced cost predicts the objective slope; the ratio test selects a feasible replacement. A positive improving step lowers the intersection; a degenerate pivot can leave it unchanged.

Two factors determine the work

Count work per pivot and the number of pivots. Distinguish worst-case, average-case, and smoothed claims; finite termination alone does not guarantee a short run.

Starting and finishing: Phase I supplies a feasible basis or proves infeasibility. An anticycling rule supports termination while Phase II optimizes the original objective.

References and Paths for Independent Study

Connect a pivot to its picture and its cost

Interpret the basic weights, reduced-cost gap, and leaving choice in the column picture. Separate work per pivot from the number of pivots.

Textbook: Bertsimas and Tsitsiklis, *Introduction to Linear Optimization*, §§3.6–3.7.

Review: [Simplex III: Two-Phase Simplex](#) explains how to find a feasible starting basis.

Further reading: [Santos: the Hirsch counterexample](#); [Spielman and Teng: smoothed analysis](#); [Disser and Mosis: the pivot-rule complexity question](#).

Distinguish a short edge path, an improving path, and an algorithm that finds such a path efficiently.